Fragmenting Capital

Structural Change in Cross Border Venture Capital

2010–2024

Abstract. Using 1.08 million investor participations from 2010–2024, we study how China's position in the international venture-capital network changed. Spectral community estimates, quarterly change points and country-pair models distinguish a broad decline involving China in 2019 from an additional US–China bilateral decline from 2022. China's country-level community position changes relatively little, while its fitted activity share and the prevalence of US–China bridge investors fall sharply. The later bilateral decline is largest in deep-tech and industrial sectors. International participation remains high overall. The evidence supports selective separation around China, although overlapping shocks, data coverage and the small number of treated country pairs limit causal interpretation.

1 Introduction

Venture capital depends on relationships. Investors syndicate rounds, exchange information and introduce startups to other funds, so the network of investments matters alongside the amount of capital raised (Sorenson and Stuart, 2001; Hochberg et al., 2007). These relationships remain geographically concentrated: sourcing and monitoring distant startups are costly (Chen et al., 2010; Bernstein et al., 2016). They also support VC's role in financing innovation (Lerner and Nanda, 2020). As VC expanded internationally, political and regulatory changes could disrupt established connections without reducing international investment everywhere.

We ask which features of the cross-border VC network changed between 2010 and 2024, when those changes occurred, and whether they reflect a general decline in China's activity or a shift specific to US–China ties. This distinction matters for how we interpret the changes. Fewer US–China investments can arise because China attracts and supplies less international VC overall, because the two countries become less connected conditional on activity, or both.

Several events could account for the changes: restrictions on outbound investment, the 2018–19 tariff escalation, expanded investment screening, COVID-19, China's later domestic regulatory changes and restrictions on strategic technologies. Existing trade research shows large consequences of the tariff episode (Amiti et al., 2019; Fajgelbaum et al., 2020), but the geography of private investment need not respond in the same way. Because these events overlap in time, a comparison around a single chosen policy date is difficult to interpret. Appendix C.5 gives the event chronology.

We first estimate network structure, then detect breaks in its features, and finally compare activity-adjusted country-pair outcomes. The analysis finds an earlier China-wide contraction and a later bilateral decline. The main text develops that argument; the appendix contains the descriptive plots, community details, diagnostics and supplementary analyses needed to assess it.

2 Data and network construction

We use the university's licensed April 2026 PitchBook extract. The sample includes completed Seed, Early Stage VC and Later Stage VC rounds dated 2010–2024; we exclude 2025 because of reporting lags. We remove individual angels, failed investors and acquirers, and observations without an investor or startup headquarters country (1.3%). We treat mainland China separately from Hong Kong, Taiwan and Macau. The final sample contains 372,731 rounds, 1,078,311 investor participations, 89,685 investors and 207,938 startups.

A participation records one investor in one round. For annual investment networks, we collapse repeated participations into an investor–startup–year tie, yielding 1,050,067 ties. A tie is cross-border when the two headquarters countries differ. These counts measure investment relationships rather than dollars. Appendix A reports the full sample table and industry and geographic breakdowns. We use six industry groups: AI/deep tech, software/IT, healthcare/life sciences, fintech, consumer, and industrial/other.

We assign industries using PitchBook's primary group and verticals. AI/deep tech covers semiconductors, hardware, communications and the AI, quantum, robotics and autonomy verticals. Fintech includes crypto; consumer includes edtech and media; industrial/other includes energy, materials and B2B products. These groups separate sectors with different potential policy exposure, but do not map investments directly to legal restrictions. Headquarters coding also measures location rather than the nationality of a fund's managers or capital providers.

We construct three network representations. First, directed investor–startup ties describe international participation and industry composition. Second, $X_{ij,t}$ counts ties from investors in country i to startups in country j. We keep 51 countries with at least 1,000 cross-border ties over the sample, excluding five offshore jurisdictions whose headquarters codes often reflect fund domicile: the Cayman Islands, British Virgin Islands, Seychelles, Bermuda and Panama. For community analysis, we symmetrise cross-border flows as $Y = (X + X^\top)/2$, normalise to $Z = D^{-1/2} Y D^{-1/2}$ with $D = \mathrm{diag}(Y\mathbf{1})$, and retain edges whose Z_{ij} exceeds its mean. Directed counts are retained for the later flow models.

Third, we construct investor co-investment graphs over three-year windows. Investors must participate in at least ten rounds, and an edge requires co-investment in at least two rounds. We analyse the giant component: 3,279 nodes and 23,253 edges in 2015–17, versus 8,370 nodes and 93,711 edges in 2022–24. Country graphs capture broad geographic structure; investor graphs reveal syndication relationships within and across countries. Thresholding and giant-component selection therefore define which relationships the structural results describe.

The descriptive evidence already suggests a distinction between global integration and China's experience. The cross-border share rises from 19% in 2010 to 34% in 2021 and remains above 28% afterwards. US–China ties, however, fall from 6.2% of cross-border ties in 2018 to 3.4% in 2019 and 1.1% in 2024. China's ties with other countries fall from 10.3% to 6.1% and then 2.7%. Bilateral counts briefly recover during the 2021 boom, while shares do not. The plots in the appendix show why declining shares and temporarily recovering levels can coexist.

3 Statistical models and spectral estimation

For an undirected adjacency matrix A, the degree-corrected mixed-membership model (DCMM) assumes independent edges with

$$\Pr(A_{ij} = 1) = \theta_i \theta_j \boldsymbol{\pi}_i^\top P \boldsymbol{\pi}_j, \qquad i \neq j. \tag{1}$$

The activity parameter $\theta_i > 0$ allows heterogeneous degrees. The nonnegative membership vector $\boldsymbol{\pi}_i \in \mathbb{R}^K$ sums to one; P is symmetric and full-rank, with unit diagonal. This combines degree correction and mixed membership (Karrer and Newman, 2011; Airoldi et al., 2008; Jin et al., 2024), relaxing the stochastic block model (Holland et al., 1983). Pure memberships give the degree-corrected block model; equal activity parameters give a mixed-membership model. DCMM distinguishes three possible changes: lower activity, a different community position, and weaker affinity between communities.

We estimate communities using SCORE eigenvector ratios (Jin, 2015). Investor graphs first use the SCORE+ normalisation $L_\tau = H^{-1/2} A H^{-1/2}$, where $H = \operatorname{diag}(A\mathbf{1}) + \bar{d} I$ (Jin et al., 2021). After ordering eigenvectors by eigenvalue magnitude, we form

$$\hat{R}_i = \left(\frac{\hat{\xi}_2(i)}{\hat{\xi}_1(i)}, \ldots, \frac{\hat{\xi}_K(i)}{\hat{\xi}_1(i)} \right),$$

clipping entries to $[-\log n, \log n]$. The ratios reduce degree effects and place rows near a simplex. Vertex hunting smooths over $n/50$ nearest neighbours, uses a $10K$-centre k-means sketch, and applies successive projection; the country fit uses a maximum-volume simplex. Negative barycentric coordinates $\hat{w}_i$ are truncated and renormalised. Mixed-SCORE sets $\hat{\pi}_i(k) \propto \hat{w}_i(k)/\hat{b}_1(k)$, normalising to sum to one, with

$$\hat{b}_1(k) = \left[\hat{\lambda}_1 + \hat{v}_k^\top \operatorname{diag}(\hat{\lambda}_2, \ldots, \hat{\lambda}_K) \hat{v}_k \right]^{-1/2}.$$

Here $\hat{v}_k$ is the estimated simplex vertex. Country scree plots motivate $K = 3$; the largest investor eigen-ratio over $k = 4, \ldots, 10$ selects $K = 4$ in 2015–17 and $K = 8$ in 2022–24. Appendix B gives the scree and degree plots; Appendix C records the algorithm.

The full-period country graph has 304 edges and separates an Atlantic core, Asia-Pacific, and an emerging/peripheral group. The US is highly mixed, with weights $(0.38, 0.34, 0.28)$, while China lies near Asia-Pacific. Across eleven five-year windows matched by the Hungarian algorithm, China's Asia-Pacific weight falls from 0.84 to 0.70; Singapore's Atlantic weight rises from 0.23 to 0.36. Most positions move modestly (Appendix B.1). A normalised summary $C_{k\ell} = B_{k\ell}/\sqrt{B_{kk} B_{\ell\ell}}$, $B = \hat{\Pi}^\top A \hat{\Pi}$, describes connectivity; it is distinct from a general estimator of the model's P.

4 Investor communities and bridges

Investor communities separate by both geography and industry. The 2015–17 fit has four communities: China; Japan–Korea–Singapore; transatlantic healthcare; and a large US-led group. By 2022–24, the eight-community fit distinguishes national groups in China,

Japan and Korea, a US core, a healthcare group, and additional communities involving fintech and crypto investors. Appendix B.3 contains the full community table, membership distributions and network drawings.

China's community becomes more internally concentrated. Its China-HQ share rises from 86% to 93%, while its internal edge share rises from 84% to 92%. Average normalised connectivity with other communities falls from 0.38 to 0.18. Among US-HQ investors, the share with estimated China membership above 0.2 falls from 5.9% to 1.0%, and mean China membership falls from 0.094 to 0.054. China-HQ investors' mean own-community weight rises from 0.73 to 0.82. Because K changes across the two fits, these comparisons describe the estimated partitions rather than a change under a fixed community definition.

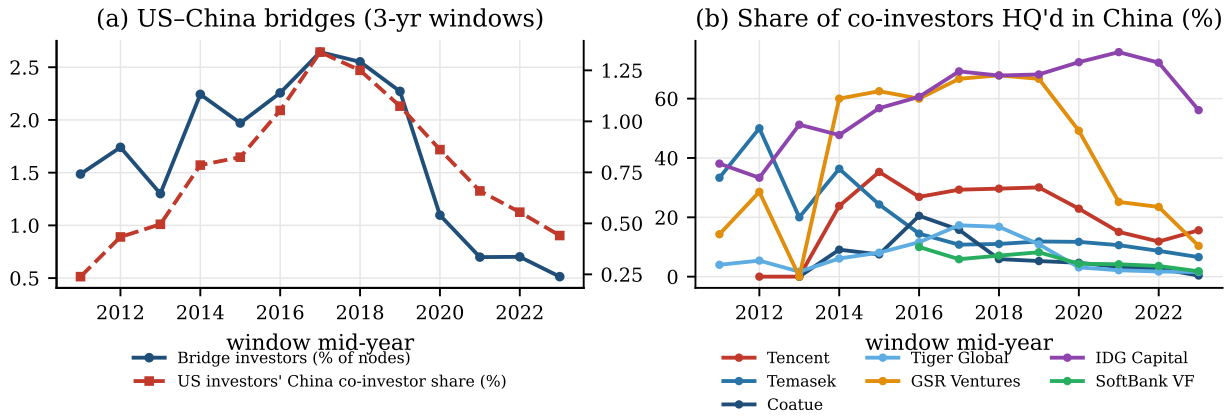


Figure 1: Left: investors bridging US-HQ and China-HQ co-investors, and US investors' mean China-HQ co-investor share, in three-year windows. Right: China-HQ co-investor shares of selected crossover investors. Window dates are midpoints.

A measure based on fixed headquarters provides a more comparable view. We define a bridge as an investor whose neighbours are at least 20% US-HQ and at least 20% China-HQ. Bridges represent 2.0–2.6% of nodes in windows centred on 2014–19, falling to 1.1% in 2020, 0.7% in 2021–22 and 0.5% in 2023. US investors' mean share of China-HQ co-investors falls from 1.3% at its 2017–18 peak to 0.4%.

The examples of named investors are consistent with that broader contraction: several crossover investors reduce their China-HQ co-investor shares, while dedicated China funds remain concentrated within the China community. These patterns describe fewer connections between the two investor populations, not complete disconnection. An additional investor-level regression in Appendix C.4 examines pre-existing China exposure, with an explicit caveat about mean reversion.

5 Change points in quarterly network features

We compute features on trailing four-quarter windows from 2010Q4 to 2024Q4 (57 observations). China features are bilateral observed-to-expected affinity, China's cross-border strength share, log US–China ties, and headquarters-based China–US and China–other connectivity. Global features exclude China: the cross-border share, the normalised coun-

try network's second eigenvalue, and US–other connectivity. Headquarters groups keep labels consistent across windows.

After standardisation, we use multivariate binary segmentation with a Gaussian mean-shift cost in *ruptures* (Truong et al., 2020). BIC uses an effective sample size of $n/4$. Stability checks use 300 moving-block bootstrap replications of within-segment residuals, block length four, and PELT with an RBF cost (Killick et al., 2012). Appendix C.3 reports alternative samples and features.

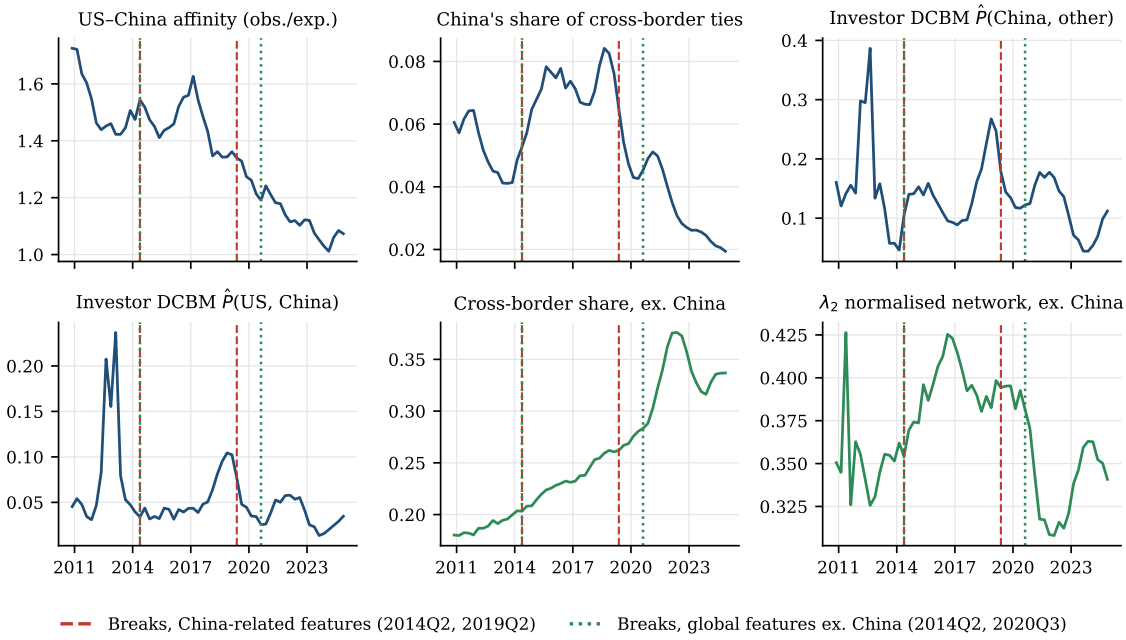


Figure 2: Quarterly features. Red dashed lines mark China-group breaks; green dotted lines mark global breaks excluding China. Dates end trailing annual windows. Small early investor networks produce noisy series.

China-related features select 2014Q2 and 2019Q2; global features select 2014Q2 and 2020Q3. The best single breaks are 2019Q2 for China-related features and 2020Q3 for global features. Country-only and post-2013 specifications locate the main China change in 2019Q1–Q2 and the global change in 2020Q2–Q3. A later China break appears around 2022Q4–2023Q1 when additional changes are allowed, so it is less stable than the main break.

The China shift precedes the global shift by about five quarters. Its 2019Q2 window covers 2018Q3–2019Q2, overlapping tariffs, investment screening and China's funding contraction. It predates COVID but cannot distinguish the earlier events. Overlap blurs dates, and $n/4$ is heuristic. Breaks detect feature changes rather than individual policy shocks.

6 Activity and a conditional flow benchmark

To distinguish lower activity from weaker bilateral affinity, we fit a Poisson extension of the country model to directed cross-border counts:

$$\mathbb{E}[X_{ij,t}] = a_{i,t} b_{j,t} \boldsymbol{\pi}_i^\top P \boldsymbol{\pi}_j. \tag{2}$$

Memberships are fixed at the full-period Mixed-SCORE estimates. The affinity matrix P is fitted on 2014–17 using multiplicative updates, with $P_{11} = 1$. Annual $a_{i,t}$ and $b_{j,t}$ are fitted by iterative proportional fitting to match realised outgoing and incoming country totals, excluding domestic cells. We use 60 outer iterations and 50 inner fitting rounds; Appendix C gives the updates. Pre-period correlation between $\log(1+X)$ and $\log(1+\hat{X})$ is 0.87, which describes the in-sample fit.

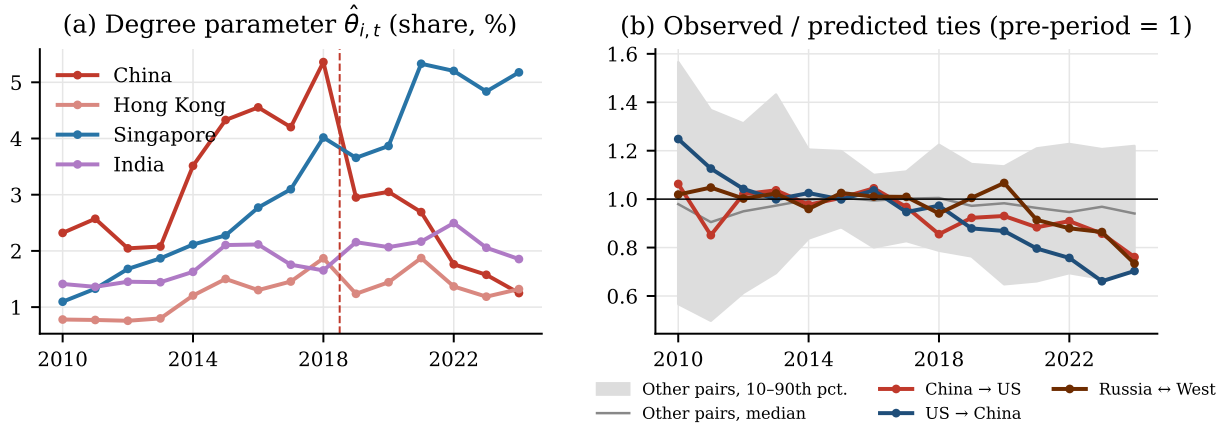


Figure 3: Left: activity parameters $\hat{\theta}_{i,t} = \sqrt{a_{i,t} b_{i,t}}$ as shares of their total. Right: observed/predicted ties, normalised to one in 2014–17. The band spans the 10th–90th percentiles across 78 large country pairs.

China's fitted activity share rises from 2.0% in 2012 to 5.4% in 2018, falls to 3.0% in 2019 and reaches 1.3% in 2024. Singapore overtakes China in 2019. This is a parameter share rather than China's observed share of ties; its sharp fall is consistent with a broad contraction involving China. It also contrasts with the smaller movements in China's estimated community position.

Conditional bilateral weakness develops more gradually. The normalised ratio of observed to predicted US investment ties in China is 0.88 in 2019, 0.80 in 2021 and 0.66–0.70 in 2023–24. For China investment in the US, it is 0.92 in 2019 and 0.76 in 2024. The US-to-China ratio reaches roughly the 10th percentile of large pairs by 2023–24; the China-to-US ratio is in the bottom quartile in 2024. Russia–West reaches 0.73 in 2024.

This exercise separates activity from residual bilateral weakness under a fitted structure. It is a conditional benchmark rather than an untreated policy counterfactual: annual margins are observed after the changes, and memberships use the full period. Holding memberships fixed also prevents it from independently measuring all three DCMM change channels. Within those limits, it suggests that broad activity loss explains much of the 2019 decline, with further bilateral weakness later.

7 Country pair comparisons

We estimate a gravity Poisson pseudo-maximum-likelihood model (Anderson and van Wincoop, 2003; Santos Silva and Tenreyro, 2006) for annual directed country flows:

$$\mathbb{E}[X_{ij,t}] = \exp\left(\alpha_{it} + \gamma_{jt} + \delta_{ij} + \sum_{s \neq 2017} \beta_s^{UC} U_{ij} \mathbf{1}[t = s] + \sum_{s \neq 2017} \beta_s^{RW} W_{ij} \mathbf{1}[t = s]\right). \quad (3)$$

Here U_{ij} marks the two US–China directions and W_{ij} the Russia–West pairs. Investor-country–year and startup-country–year effects absorb country-wide changes; pair effects absorb stable affinity. The coefficients measure pair-specific deviations relative to 2017, conditional on these controls.

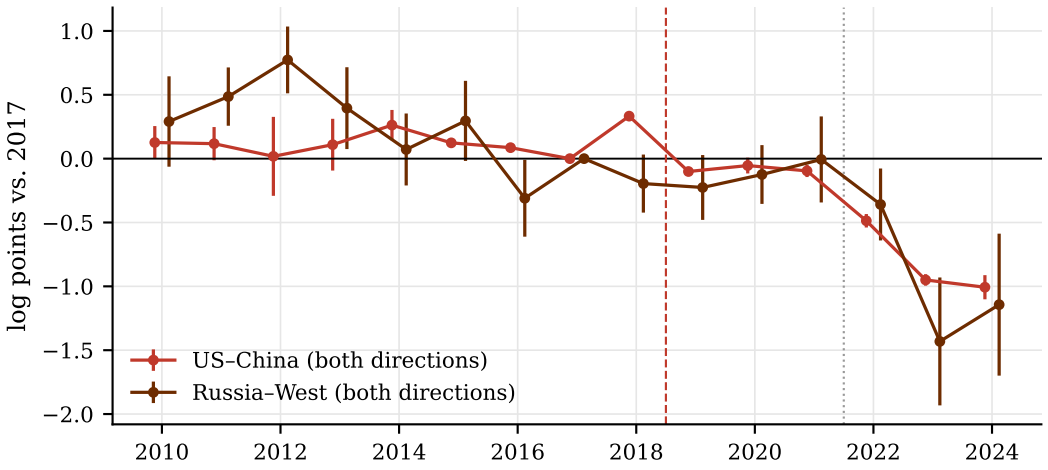


Figure 4: Gravity event study relative to 2017. Dyad-clustered 95% intervals are illustrative because US–China has only two treated directions. Russia–West provides a contextual comparison rather than a causal control group.

US–China coefficients are broadly flat in 2014–17, rise by 0.33 in 2018, and lie between −0.05 and −0.10 in 2019–21. They fall to −0.49 in 2022, −0.95 in 2023 and −1.01 in 2024: approximately 39%, 61% and 63% fewer ties relative to the conditional 2017 baseline. Russia–West also falls after 2021, with coefficients of −0.36 in 2022 and −1.43 in 2023. Similar timing alone does not establish a common mechanism.

With only two treated directions, we supplement the event study with a descriptive placebo comparison. We fit a separate post-period model with the same fixed effects for each of 78 pairs averaging at least 20 annual ties in 2014–17. The two US–China directions rank 12th and 14th for declines in 2019–21. In 2022–24, China-to-US ranks second and US-to-China fifth. These ranks show that the later decline is unusually large; without exchangeability, they are not permutation p-values.

Together, the two flow models distinguish the earlier broad China-related decline from the later bilateral decline. Country-year effects absorb much of the 2019 change, while unusually weak US–China links emerge after 2021. Those controls cannot separate underlying activity from uniform coverage changes or identify which overlapping geopolitical and domestic shocks caused the differences.

8 Industry differences and discussion

The industry PPML model adds pair–industry effects and investor- and startup-country–industry–year effects. US–China–industry–period indicators compare 2019–21 and 2022–24 with 2010–17, omitting 2018. Each coefficient compares the bilateral change within an industry with changes in the same industry among other pairs.

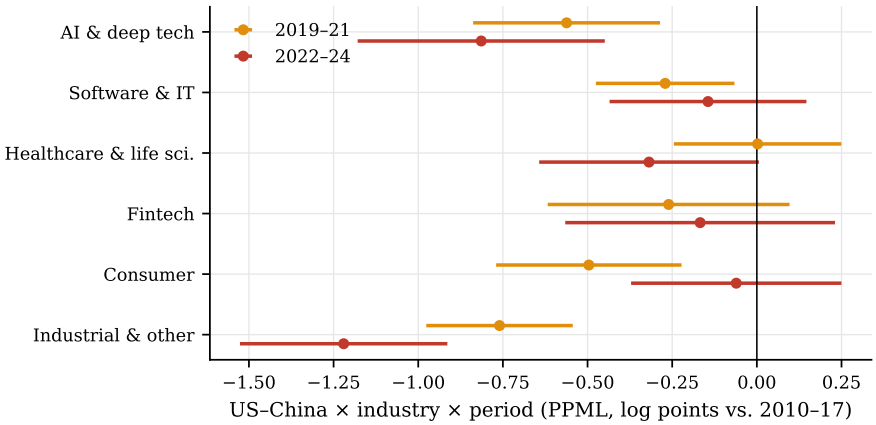


Figure 5: US–China industry-period coefficients in log points. The plotted 95% intervals use pair–industry clustering; few treated dyads limit conventional inference.

The largest persistent declines are in AI/deep tech (−0.56, then −0.81) and industrial/other (−0.76, then −1.22). Consumer ties fall initially (−0.50), while software's decline is smaller (−0.27). Healthcare weakens later (−0.32 in 2022–24). Concentration in strategic sectors is consistent with policy-related disruption, but broad industry codes cannot establish which investments faced particular restrictions.

The evidence points to selective separation around China: a broad activity decline in 2019, followed by additional bilateral weakness from 2022. Bridge investors become less common even as aggregate cross-border participation remains high. Country membership positions alone would miss much of that change.

Several limits matter. The analysis is observational, with only two treated US–China directions. The 2019 window combines domestic funding changes and international restrictions; annual 2022 estimates cannot isolate late-year export controls, and later rules cannot explain an earlier onset. PitchBook coverage and headquarters relocations may change measured country flows. Thresholds, vertex hunting and changing K affect the investor fits. Overlapping windows induce dependence, and the residual bootstrap omits uncertainty in estimated network features.

A next step is to estimate memberships consistently over time and map startup technologies to policy exposure more precisely. The appendix includes additional investor regressions, diagnostics and implementation details, along with all supporting plots. Replication requires the licensed PitchBook data and analysis scripts.

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A Sample and descriptive evidence

A.1 Sample composition and aggregate activity

Table A1 reports the sample by period. Participations are round-level records, while ties are deduplicated investor–startup–year pairs. Counts of distinct investors and startups overlap across periods, so their period totals should not be added.

Table A1: Sample composition by period

Period	Rounds	Ties	Investors	Startups	Cross-border share	Investor countries	Startup countries
2010–13	36,520	82,981	14,903	27,609	19.9%	103	115
2014–17	81,763	196,379	31,992	59,009	21.8%	133	147
2018–21	136,579	404,762	52,869	94,751	29.5%	156	170
2022–24	117,869	365,945	50,708	93,310	31.3%	160	167

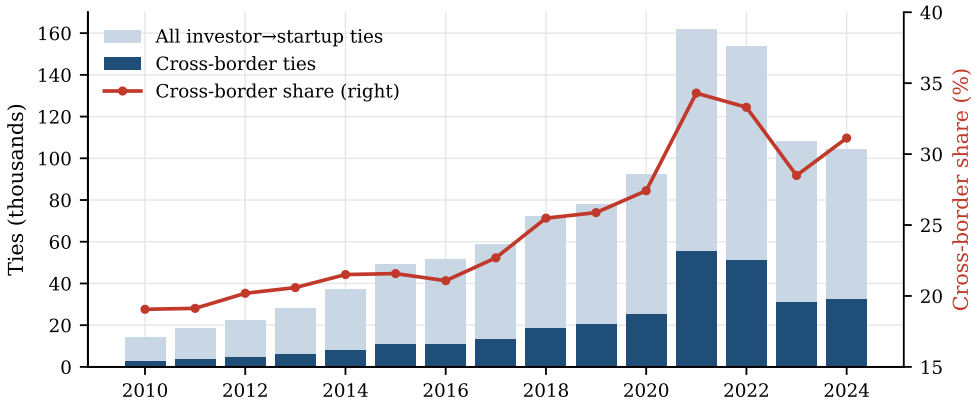


Figure A1: Global VC activity and the cross-border share of investor–startup ties, 2010–2024. Activity grows more than tenfold to its 2021 peak and then contracts by about one third.

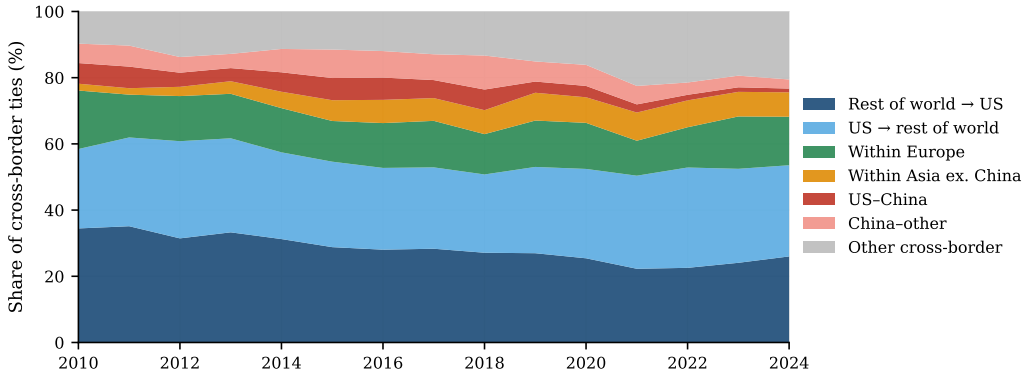


Figure A2: Cross-border ties by region pair. China-related ties peak at 16.5% in 2018 and fall to 3.8% in 2024, despite rising aggregate international participation.

A.2 Bilateral ties and industry composition

US–China ties account for 5.4–6.2% of cross-border ties in 2014–18, falling to 3.4% in 2019 and 1.1% in 2024. China's ties with other countries fall from 10.3% in 2018 to 6.1% in 2019 and 2.7% in 2024. Counts temporarily recover in 2021, but their shares do not. This distinction is relevant when interpreting the global VC boom.

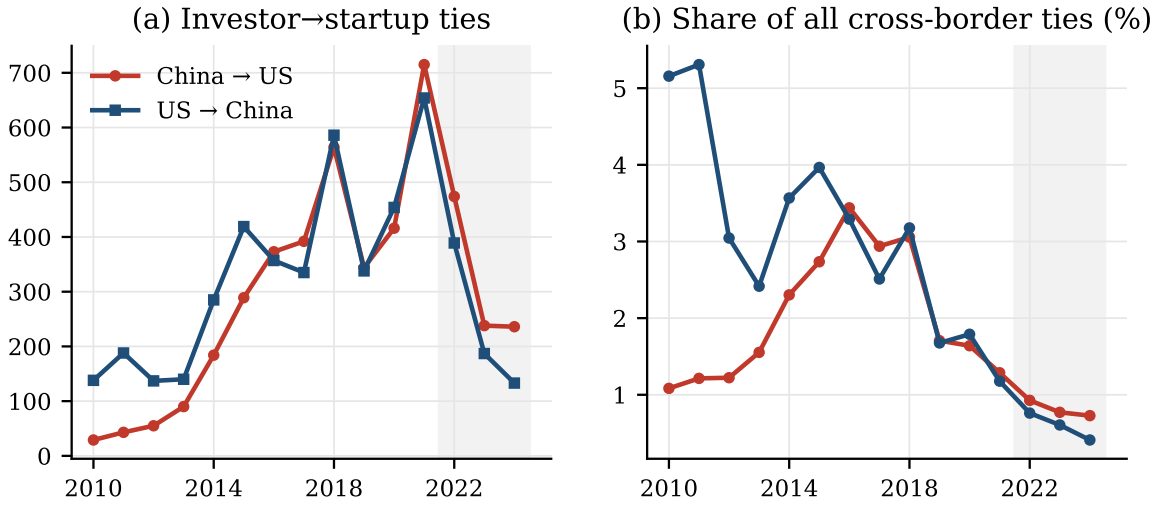


Figure A3: US–China investment ties in levels and as a share of all cross-border ties. Shading marks 2022–24. A temporary rise in counts during the 2021 boom does not restore the earlier share.

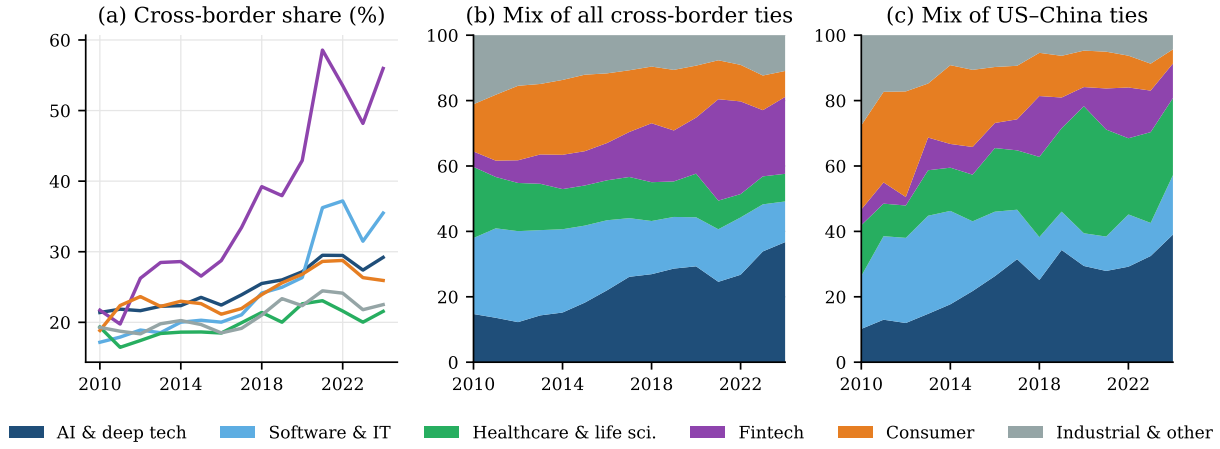


Figure A4: Industry composition and international participation. Panels show cross-border shares by industry, the mix of all cross-border ties, and the mix of US–China ties.

The cross-border share rises in all six industry groups. Fintech increases from 22% in 2010 to 59% in 2021, reflecting the highly international nature of many crypto rounds. Within US–China ties, AI and deep tech rise from 10% in 2010 to 32% in 2017 and 39% in 2024. Consumer falls from 26% to 4%. These are composition shares: a sector can gain share while its number of bilateral ties falls.

A.3 Country specialisation and degree heterogeneity

Revealed comparative advantage (RCA) compares an industry's share of a country's inbound foreign ties with that industry's share worldwide. A value of one indicates the same mix as the world. In 2016–24, China's inbound foreign VC is relatively concentrated in healthcare (RCA 1.7) and consumer (1.5). Other patterns include deep tech in Israel, consumer in India and Korea, fintech in Singapore, Hong Kong and the UAE, and healthcare in Belgium, Ireland and Denmark.

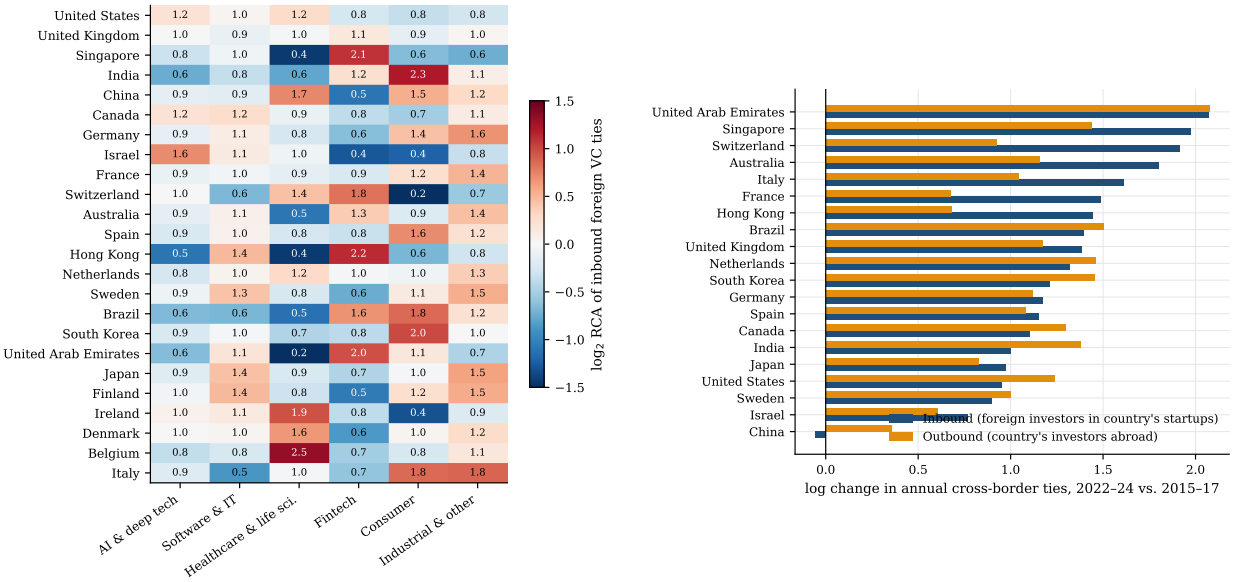


Figure A5: Left: inbound foreign-VC RCA for the 24 largest destinations, 2016–24. Right: log change in annual cross-border ties, 2022–24 relative to 2015–17. China has the weakest inbound change among the large countries shown; its outbound log growth is 0.35.

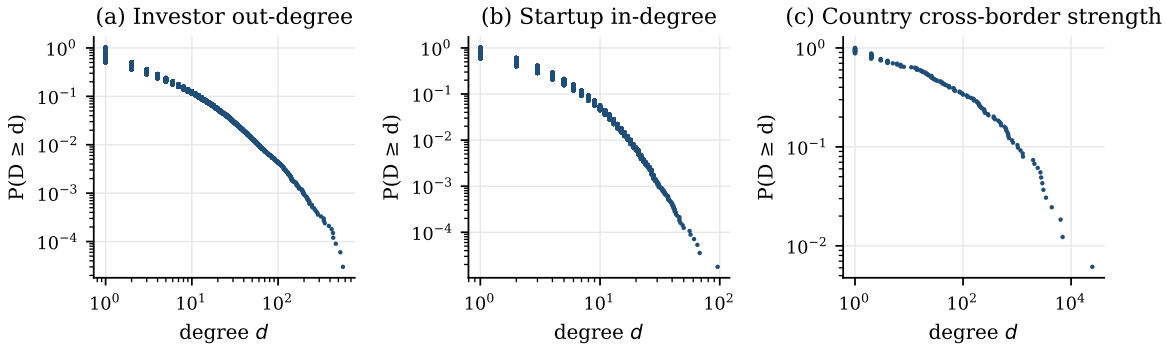


Figure A6: Empirical complementary cumulative degree distributions in 2016–18 on log–log axes. Investor out-degrees, startup in-degrees and country strengths are highly unequal.

The US accounts for about 40% of cross-border strength. A small number of investors, including accelerators and active seed funds, also account for many ties. This heterogeneity motivates degree correction before interpreting spectral communities.

B Spectral diagnostics and network structure

B.1 Country memberships and model dimension

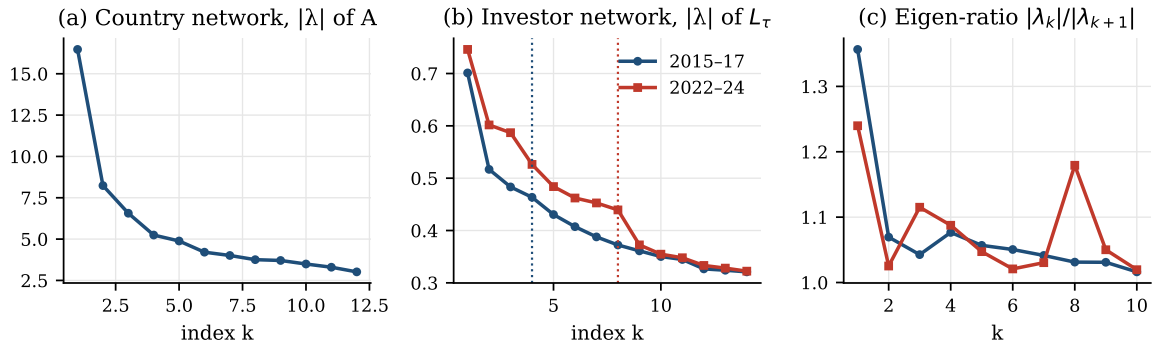


Figure A7: Scree diagnostics for the country and investor networks. Country gaps after two and three eigenvalues motivate $K = 3$; investor eigen-ratios select $K = 4$ in 2015–17 and $K = 8$ in 2022–24.

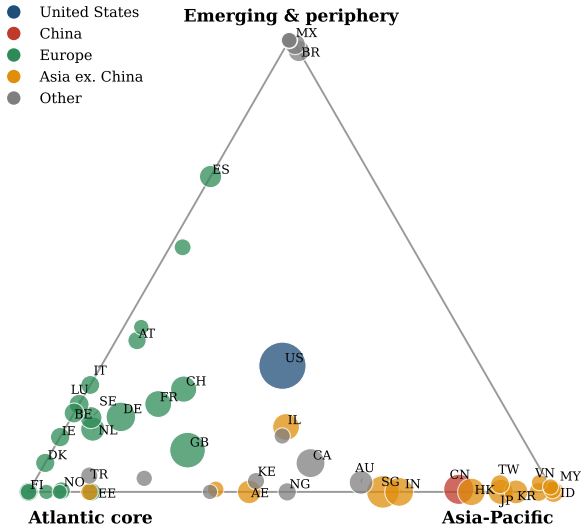


Figure A8: Mixed-SCORE memberships in the 51-country graph, 2010–24. The three vertices describe an Atlantic core, Asia-Pacific, and an emerging/peripheral group. Marker size reflects cross-border strength.

The US has weights $(.38, .34, .28)$ and Israel $(.44, .42, .14)$. Atlantic and Asia weights are $(.33, .67)$ for Singapore and $(.58, .42)$ for the UAE. China is $(.18, .82, .01)$ and Hong Kong $(.16, .84, .00)$, up to rounding. The normalised connectivity index, ordered Atlantic, Asia and periphery, is

$$\hat{C} = \begin{pmatrix} 1 & .50 & .51 \\ .50 & 1 & .24 \\ .51 & .24 & 1 \end{pmatrix}.$$

This index summarises the fitted membership-weighted adjacency; it should be distinguished from the block matrix estimated in the directed Poisson flow model.

B.2 Country trajectories and investor network drawings

Across eleven five-year windows, communities are matched to the full-period country vertices by the Hungarian algorithm. Singapore's Atlantic weight rises from .23 in 2010–14 to .36 in 2020–24. China's Asia-Pacific weight falls from .84 to .70, while the US remains near the centre. These movements are modest compared with the changes in China's activity.

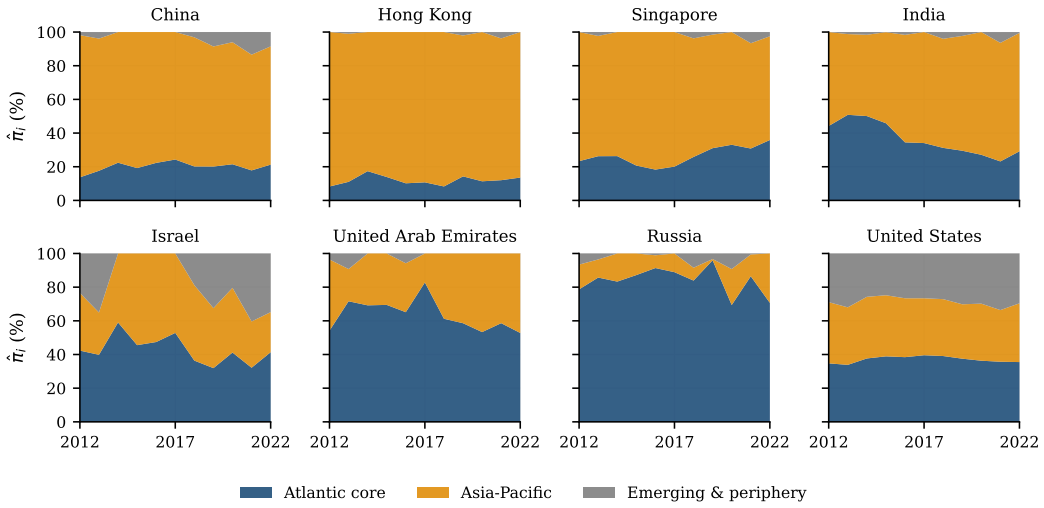


Figure A9: Country membership trajectories over five-year rolling windows, labelled by mid-year. Labels are matched to the full-period country solution.

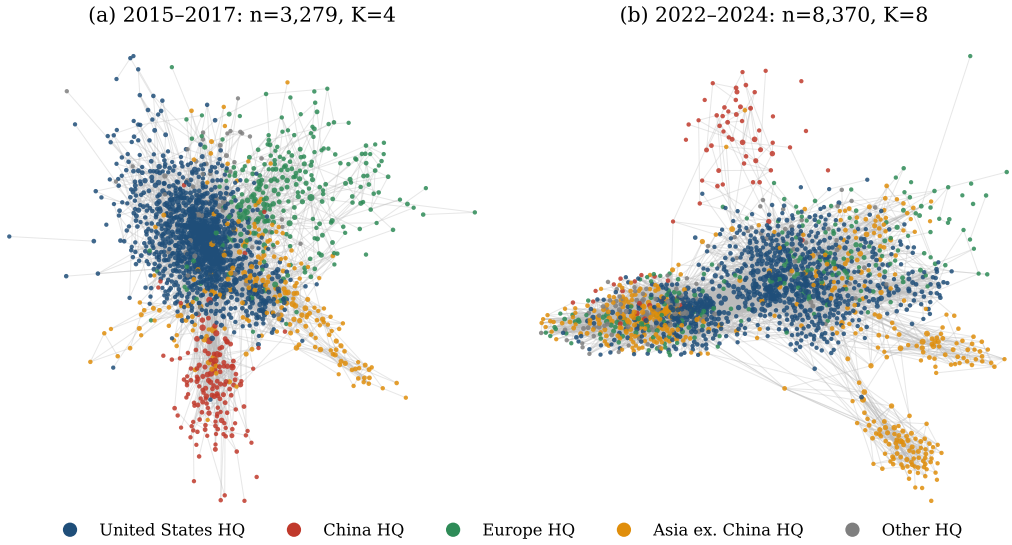


Figure A10: Investor network drawings for 2015–17 and 2022–24. Only the 1,600 highest-degree investors and a random 6,000 edges are displayed; colour denotes headquarters region. Separation in this sampled drawing does not imply that a group is disconnected in the full giant component.

B.3 Complete investor community summaries

Hard assignments use $\arg\max_k \hat{\pi}_i(k)$. The internal column is the share of a community's co-investment edges that remains within it. Communities vary in both geography and portfolio industry; the different selected values of K limit direct comparisons of individual labels across eras.

Table A2: Investor communities in 2015–17 and 2022–24

ID	Size	Leading HQ countries	Most-connected investors	Internal
2015–17: 3,279 nodes and 23,253 edges				
C1	533	China 86%, US 5%, HK 4%	IDG Capital, ZhenFund, HongShan, MPC	.84
C2	324	Japan 27%, Korea 23%, Singapore 12%	DCM, Wavemaker, East Ventures, Peak XV	.73
C3	447	US 45%, UK 14%, France 13%	Bpifrance, Fidelity, Canaan, Venrock; healthcare 47%	.72
C4	1,975	US 63%, UK 6%, Germany 4%	500 Global, NEA, Y Combinator	.93
2022–24: 8,370 nodes and 93,711 edges				
C1	905	China 93%, HK 3%, US 2%	HongShan, Suzhou Oriza, Shenzhen Capital, CICC Capital	.92
C2	332	Japan 83%, US 9%	Global Brain, MUFG Capital, SMBC VC	.82
C3	353	Korea 94%	KB Investment, IBK, Korea Investment Partners	.94
C4	4,194	US 59%, UK 6%, Germany 4%	Gaingels, Alumni Ventures, Plug and Play, a16z	.84
C5	480	US 54%, France 13%, Netherlands 6%	GV, NEA, Bpifrance; healthcare 52%	.63
C6	819	US 45%, India 28%	Coinbase Ventures, Samsung NEXT	.52
C7	924	US 17%, UK 14%, Singapore 11%	Shima Capital, MH Ventures	.55
C8	363	US 30%, Singapore 17%, HK 7%	Animoca Brands, Big Brain, GSR; fintech 51%	.45

Normalised mutual information between community assignments and HQ country is .34 in 2015–17 and .35 in 2022–24; modularity is .33 and .46. The average China-to-other-community connectivity index falls from .38 to .18. These summaries describe the fitted partitions rather than supply a test of increasing fragmentation.

C Implementation and supplementary analyses

C.1 Mixed-SCORE computation

The investor-network input is the adjacency matrix A on its giant component and a selected community count K. The computation follows these steps (Jin, 2015; Jin et al., 2021, 2024):

1. Compute $d = A\mathbf{1}$, $H = \mathrm{diag}(d) + \bar{d}I$ and the regularised matrix $L_\tau = H^{-1/2}AH^{-1/2}$.
2. Extract the leading K eigenpairs $(\hat{\lambda}_k, \hat{\xi}_k)$ in decreasing absolute eigenvalue order, choosing $\hat{\xi}_1 > 0$ on the connected component.
3. Set $\hat{R}_{ik} = \hat{\xi}_{k+1}(i)/\hat{\xi}_1(i)$ for $k = 1, \ldots, K-1$ and clip to $[-\log n, \log n]$.
4. Replace each ratio row by its mean over $n/50$ nearest neighbours. Run k-means with $10K$ centres, then successive projection on the centres to obtain vertices $\hat{v}_1, \ldots, \hat{v}_K$ (Araújo et al., 2001). For the country graph, use the maximum-volume simplex among the centres.
5. Solve $\sum_k \hat{w}_i(k) = 1$ and $\sum_k \hat{w}_i(k)\hat{v}_k = \hat{R}_i$. Set negative coordinates to zero and renormalise.
6. Compute $\hat{b}_1(k) = [\hat{\lambda}_1 + \hat{v}_k^\top \mathrm{diag}(\hat{\lambda}_2, \ldots, \hat{\lambda}_K)\hat{v}_k]^{-1/2}$. Divide $\hat{w}_i(k)$ by $\hat{b}_1(k)$ and normalise to obtain $\hat{\pi}_i$.
7. For descriptive connectivity, form $B = \hat{\Pi}^\top A \hat{\Pi}$ and $C_{k\ell} = B_{k\ell}/\sqrt{B_{kk}B_{\ell\ell}}$. Using HQ labels yields the US/China/other indices used in quarterly features.

C.2 Directed Poisson fitting

With country memberships fixed at their full-period values, the 2014–17 affinity fit uses multiplicative updates

$$P_{k\ell} \leftarrow P_{k\ell} \frac{\sum_t (\Pi^\top X_t \Pi)_{k\ell}}{\sum_t (\Pi^\top \hat{X}_t \Pi)_{k\ell}},$$

normalised to $P_{11} = 1$, with 60 outer iterations. Each inner step runs 50 rounds of iterative proportional fitting for annual origin and destination activity terms, excluding diagonal cells. Full-period memberships and realised annual margins make these predictions conditional benchmarks. The fit does not estimate the network that would have prevailed without a particular policy.

C.3 Change-point sensitivity and investor memberships

Table A3: Change-point estimates under alternative feature sets

Feature set	BIC-selected breaks	Best single
China-related	2014Q2, 2019Q2	2019Q2
Global excluding China	2014Q2, 2020Q3	2020Q3
China, country-only	2014Q2, 2019Q2, 2023Q1	2019Q2
China, from 2013Q1	2014Q1, 2017Q4, 2019Q1, 2022Q4	2019Q1
Global ex. China, country-only	2014Q2, 2020Q3	2020Q3
Global ex. China, from 2013Q1	2015Q2, 2020Q2	2020Q2

Dates end trailing four-quarter windows. Binary segmentation uses standardised series and an ℓ_2 cost; BIC uses the heuristic effective sample size $n/4$. The main China change remains in early 2019 when the small early investor networks are excluded. Later breaks depend more strongly on specification.

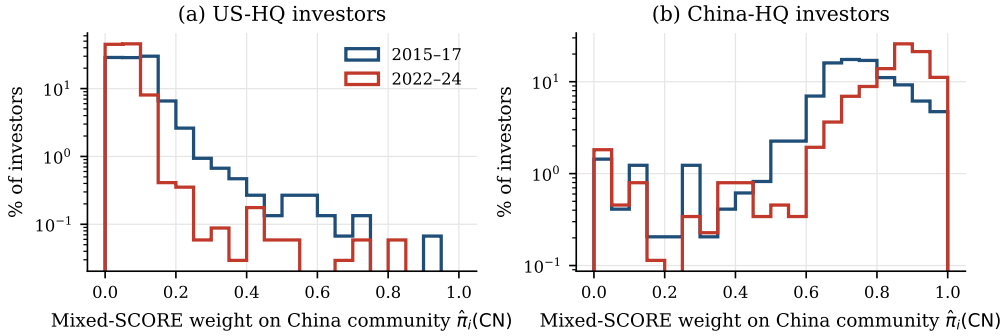


Figure A11: China-community membership distributions for US-HQ and China-HQ investors. The log scale makes intermediate weights visible. Comparisons across eras are descriptive because the selected K changes.

US investors' mean China weight falls from .094 to .054; China-HQ investors' mean own-community weight rises from .73 to .82. US investors' mean China-HQ co-investor share peaks at 1.3% in 2017–18 and falls to .4%. Selected crossover investors reduce their China-HQ co-investor shares after 2019; GSR Ventures falls from 67% to 10%.

C.4 Investor exposure regression

We follow 2,792 non-China investors active in 2015–17 and a later window; 101 initially have at least 10% China-HQ co-investors. With investor and window effects, this exposed group's share falls relative to unexposed investors by 12.7 percentage points (s.e. 2.6) in 2019–21 and 16.9 points (2.8) in 2022–24. Among exposed investors, US-HQ investors decline a further 9.6 points (4.9) relative to non-US investors by 2019–21. Baseline-outcome selection induces mean reversion; the within-exposed contrast is imprecise.

C.5 Historical context and possible extensions

The chronology below provides context for the estimated dates. Multiple events overlap the annual trailing windows, and their proximity to a detected break does not establish causation. In particular, the August 2023 order cannot explain a change that begins in 2022 or a break dated 2023Q1.

Table A4: Events relevant to the interpretation of the network changes

Date	Event
2014–15	Rapid expansion of Chinese outbound VC and of US funds' activity in China.
August 2017	Chinese restrictions on outbound investment.
April 2018	Asset-management rules tighten RMB fundraising, during the 2018–19 funding contraction.
July–September 2018	Section 301 tariff lists 1–3.
August/November 2018	FIRRMA enacted; CFIUS pilot programme for critical technologies.
March 2020	COVID-19 pandemic and a shift toward remote due diligence.
2021	Chinese platform and education regulation; global VC boom.
February 2022	Sanctions on Russia.
October 2022	US controls on advanced computing and semiconductor exports.
August 2023	US executive order on outbound investment in sensitive technologies.

The country-pair comparison uses the 78 cross-border pairs among the 51 countries averaging at least 20 annual ties in 2014–17. Each comparison coefficient is estimated in a separate PPML regression with the same fixed effects and only that pair treated. This supplies descriptive ranks; validity as formal permutation inference would require additional exchangeability assumptions.

Several extensions would sharpen the analysis. A dynamic DCMM could stabilise investor trajectories across windows, while holding the community definition fixed would improve comparisons over time. A SgnQ test of $K = 1$ within the China community could assess whether its coarse summary conceals subcommunities. Coauthorship and citation applications illustrate the flexibility of network methods (Ji and Jin, 2016); here, portfolio attributes could complement co-investment edges. Mapping startups to the 27 FIRRMA pilot industries and the technologies in the 2023 outbound-investment framework would allow more specific exposure comparisons. An LP–fund network could help distinguish genuine withdrawal from rerouting through offshore vehicles. These are proposed analyses, not results established here.